The Möbius-Screw Electron: Framed Unknot Geometry for $g = 2$ and a Capacitance Model of α

Companion note to the Flag Condensate phase-slip programme

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Sovereign-Stack ACS Research Program

July 22, 2026

Abstract

We present a geometric model of the electron as a *framed unknot*: the centerline of a $(2, 1)$-torus embedding on a torus of major radius R and minor radius a. Reparameterisation shows that the knot type is trivial (an unknot); the spin-1/2 and $g = 2$ content is attributed to the framing induced by the torus embedding, whose natural self-linking number is $Sl = p \cdot q = 2$. A separate capacitance estimate on the double-cover (Möbius) annulus, with self-stress matched to $m_e c^2$, returns a numerical value $\alpha^{-1} \approx 137.036$ under stated approximations. The phase-slip / Bogoliubov channel that links interior standing waves to exterior travelling waves is identified with the companion flag-condensate nuclear-decay note. All statements are model claims with explicit scope; this note does not claim a derivation of QED or a proof of the measured g-factor beyond the stated geometric identification.

Contents

1 Scope and claim discipline

This note is a geometric *design* for an electron model inside the flag-condensate programme. It is not a theorem of QED, not a measurement, and not a uniqueness claim.

- **In scope:** centerline geometry; torus-knot winding numbers; framing / self-linking as a candidate topological origin of $g = 2$; a capacitance estimate for α^{-1}; the link to the phase-slip transfer-matrix picture.
- **Out of scope:** radiative corrections to $g - 2$; full QED renormalisation; a first-principles derivation of R and a from the Standard Model Lagrangian; experimental claims beyond the numerical capacitance estimate under stated approximations.

2 Centerline and reparameterisation

2.1 Parametric form

Let $\varphi \in [0, 4\pi)$. The centerline is

$$\mathbf{r}(\varphi) = \begin{pmatrix} \big(R + a\cos(\varphi/2)\big)\cos\varphi \\ \big(R + a\cos(\varphi/2)\big)\sin\varphi \\ a\sin(\varphi/2) \end{pmatrix}. \tag{1}$$

2.2 Reparameterisation: $(2, 1)$-torus embedding

Set $t = \varphi/2$. As φ runs over $[0, 4\pi)$, t runs over $[0, 2\pi)$, and

$$\mathbf{r}(t) = \begin{pmatrix} (R + a\cos t)\cos(2t) \\ (R + a\cos t)\sin(2t) \\ a\sin t \end{pmatrix}, \qquad t \in [0, 2\pi). \tag{2}$$

This is a torus curve on a torus of major radius R and minor radius a. The longitude angle is $2t$ and the latitude angle is t. The winding numbers are therefore

$$p = 2 \quad \text{(longitude)}, \qquad q = 1 \quad \text{(latitude)}. \tag{3}$$

The knot type is a $(2, 1)$-torus knot.

Remark 1 (Correction of winding order). *An earlier draft labelled the object a "$(1, 2)$-torus knot." Under the standard convention that p is the longitudinal winding and q the meridional winding, the reparameterisation (2) identifies a $(2, 1)$ embedding. The distinction matters for reading the self-linking number as $p \cdot q$.*

2.3 Knot type: framed unknot

Because one winding number equals 1, the knot itself is trivial — an unknot. The spin-1/2 content in this model does not arise from a non-trivial knot (e.g. a trefoil) but from the *framing* of the unknot induced by the torus embedding.

Definition 1 (Model electron). *In this note, the electron is modelled as a* ***framed unknot****: a simple closed loop carrying two full twists of framing, equivalently a $(2, 1)$-torus embedding with self-linking number 2.*

3 Self-linking and $g = 2$

3.1 Călugăreanu identity

For a closed framed curve,

$$Sl = Tw + Wr, \tag{4}$$

where Sl is the self-linking number, Tw the total twist, and Wr the writhe [1, 2, 3].

3.2 Torus framing

A (p, q)-torus knot inherits a natural framing from the torus embedding. Under that framing the self-linking number is

$$Sl = p \cdot q = 2 \cdot 1 = 2. \tag{5}$$

3.3 Geometric identification with $g = 2$

In the Dirac theory the free-electron gyromagnetic ratio is $g = 2$ at tree level. In the present model we identify

$$g \quad \longleftrightarrow \quad Sl = 2 \tag{6}$$

as a topological bookkeeping statement: the framed unknot stores two units of linking, matching the 4π (two-turn) return of a spin-1/2 frame. This is a model identification, not a derivation of the QED vertex form factor.

Proposition 1 (Model claim)**.** *Within the framed-unknot electron model defined by* (2) *and* (5)*, the geometric self-linking number equals* 2*, and is identified with the tree-level value* $g = 2$*.*

3.4 Spin radius

Spin angular momentum of magnitude $\hbar/2$ for a circulating mass m_e at speed c on a circle of radius R fixes, in the usual order-of-magnitude estimate,

$$R = \frac{\hbar}{2m_e c}. \tag{7}$$

Equation (7) is an input scale for the geometric model, not an independent prediction.

4 Capacitance estimate for α

4.1 Double-cover annulus

Treat the Möbius / double-cover geometry as an effective annular conductor of characteristic radius R and thickness scale a. A standard approximate capacitance for a thin ring / annulus form is

$$C \approx \frac{2\pi\epsilon_0 R}{\ln(8R/a) + 1}. \tag{8}$$

(The additive constant in the denominator is scheme-dependent at this order; the logarithm dominates for $R \gg a$.)

4.2 Geometric fidelity of the capacitance model

Equation (8) is an *annulus / thin-ring* approximation to the electrostatic energy of the Möbius ribbon. The actual double-cover geometry is a twisted strip (or its immersion as a one-sided surface), not a flat annulus. Two consequences follow:

1. The logarithmic cutoff $\ln(8R/a)$ encodes an effective aspect ratio; a different conformal modulus for the true ribbon will shift the constant term and, more weakly, the argument of the logarithm.

2. The factor $e/2$ per sheet is a double-cover bookkeeping choice; alternative charge assignments change the prefactor of E_{cap} at the same order as a change of conformal modulus.

A natural revision path is to replace (8) by the capacitance of the ribbon domain under a conformal map to a standard annulus (or by a boundary-integral computation on the immersed strip), then re-solve (9). That revision is carried out as an as-implemented numerical sequel in [5] (annulus vs conformal-modulus vs Möbius BIE collocation, with aspect-ratio scan and machine-readable results). Under those revisions, moderate geometric aspect ratios yield $\alpha^{-1} = \mathcal{O}(1)$ rather than ≈ 137; the figure $\alpha^{-1} \approx 137.036$ remains a numerical output of the annulus model at a tuned cutoff, not a geometry-unique invariant of the framed unknot.

4.3 Self-stress matching

Assign charge $e/2$ to each sheet of the double cover and match the electrostatic self-energy to the rest energy:

$$E_{\text{cap}} = \frac{(e/2)^2}{2C} = m_e c^2. \tag{9}$$

Combining (8)–(9) with (7) and solving for the fine-structure combination $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ yields, under the stated approximations,

$$\alpha^{-1} \approx 137.036. \tag{10}$$

Remark 2 (Status of (10))**.** *The estimate is cutoff- and charge-split-sensitive; see §4.2. It is reported as an as-implemented estimate, not as a uniqueness theorem for α.*

5 Phase-slip unification

The companion note on nuclear decay in the flag condensate [4] treats decay as a spectral bifurcation: a confined standing-wave phase lock in the throat becomes an exterior travelling wave through Bogoliubov mode mixing across a geometric phase slip, with

$$\left|\frac{\beta}{\alpha}\right|^2 = e^{-2W} \tag{11}$$

(Gamow / WKB). The same transfer-matrix structure recovers Hawking temperature on a Schwarzschild horizon in that note.

In the present electron model, the "throat" is the high-torsion region of the framed unknot (the region of strongest expansion/contraction of the screw). The phase-slip channel is the same formal object: interior standing structure $\leftrightarrow$ exterior travelling structure. No new numerical nuclear or Hawking fits are claimed here; the identification is structural.

6 Summary of formal content

Object	Status in this note	Scope
$(2,1)$ centerline (2)	geometric definition	model
unknot + framing $Sl = 2$	topological reading	model identification
$g \leftrightarrow 2$	identification with Sl	tree-level analogy
$R = \hbar/(2m_e c)$	scale input	order-of-magnitude
C estimate (8)	approximate formula	capacitance model
$\alpha^{-1} \approx 137.036$	numerical estimate	under (9)
phase-slip / Bogoliubov	structural link	see [4]

Acknowledgments

This note consolidates the framed-unknot correction (twist, not knotting) with the capacitance and phase-slip sketches of the Flag Condensate Collaboration draft.

References

[1] G. Călugăreanu, *Sur les classes d'isotopie des noeuds tridimensionnels et leurs invariants*, Czechoslovak Mathematical Journal **11** (1961), 588–625.

[2] J. H. White, *Self-linking and the Gauss integral in higher dimensions*, American Journal of Mathematics **91** (1969), 693–728.

[3] F. B. Fuller, *The writhing number of a space curve*, Proceedings of the National Academy of Sciences **68** (1971), 815–819.

[4] Flag Condensate Collaboration, *Nuclear Decay as a Spectral Bifurcation of the Flag Condensate*, preprint (July 2026), `papers/notes/Flag_Condensate_Nuclear_Decay.tex`.

[5] Flag Condensate Collaboration, *Möbius-Ribbon Capacitance: Annulus, Conformal Modulus, and BIE Revisions of the α Estimate*, preprint (July 2026), `mobius_ribbon_capacitance.tex` / `papers/notes/Mobius_Ribbon_Capacitance.tex`.

A Interpretive framing: tornado metaphor

This appendix is interpretive only. It does not add formal claims to Sections 2–5.

Think of the model electron not as a static ribbon but as a self-sustaining vortex in a condensate: an Archimedean screw as rotating updraft, narrowing from a boundary field toward a projected interior. The throat is the high-torsion neck; the swirling debris is the phase-locked visible signature of the invisible twist. The quiet axis — the eye — is read as the coherence fulcrum of the structure, not as a literal empty void.

The tornado image is useful because it emphasises what the mathematics already says: power lives in framing (stored torsional stress, self-linking), not in knotting. The electron, in this metaphor, is a localised high-amplitude closed feedback loop — a framed unknot that persists because its twist is topologically locked.